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Of Genetics and Tectonics

The Indian subcontinent was not always located where it is today but was once attached to Africa and Madagascar. This is a relatively recent discovery. Till the early twentieth century, it was assumed that most geological features were due to vertical rather than horizontal movements in the earth's crust. Thus, the positions of the continents were considered to be essentially fixed. This was first challenged by the hypothesis of continental drift, proposed by Alfred Wegener in 1912 and expanded in his book *The Origin of Continents and Oceans* published in 1915. He suggested that the present continents once formed a single land mass that drifted apart like icebergs. The hypothesis explained an observation that had puzzled map-makers like Ortelius since the sixteenth century—the fact that land masses, especially those on the opposite sides of the Atlantic, seemed to fit together neatly like a jigsaw puzzle.

Wegener's arguments would have to wait for half a century before adequate scientific evidence could be found to support it. In the late fifties and sixties a great deal of new geological data proved that the earth's crust was a patchwork of plates that were moving relative to each other. This led to the modern theory of plate tectonics. It is still a very new field and our understanding is still evolving. Nonetheless, the broad contours of the story are now clear.

It appears that most of the earth's land mass was joined together a billion years ago in a supercontinent called Rodinia. It was probably located south of the equator but there is still a great deal of debate about its exact shape